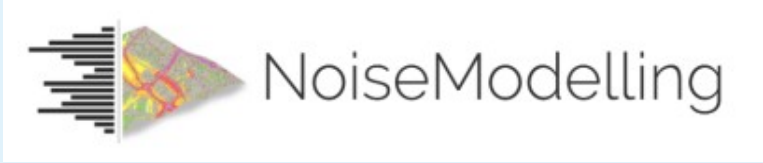




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NoiseModelling Day
2026

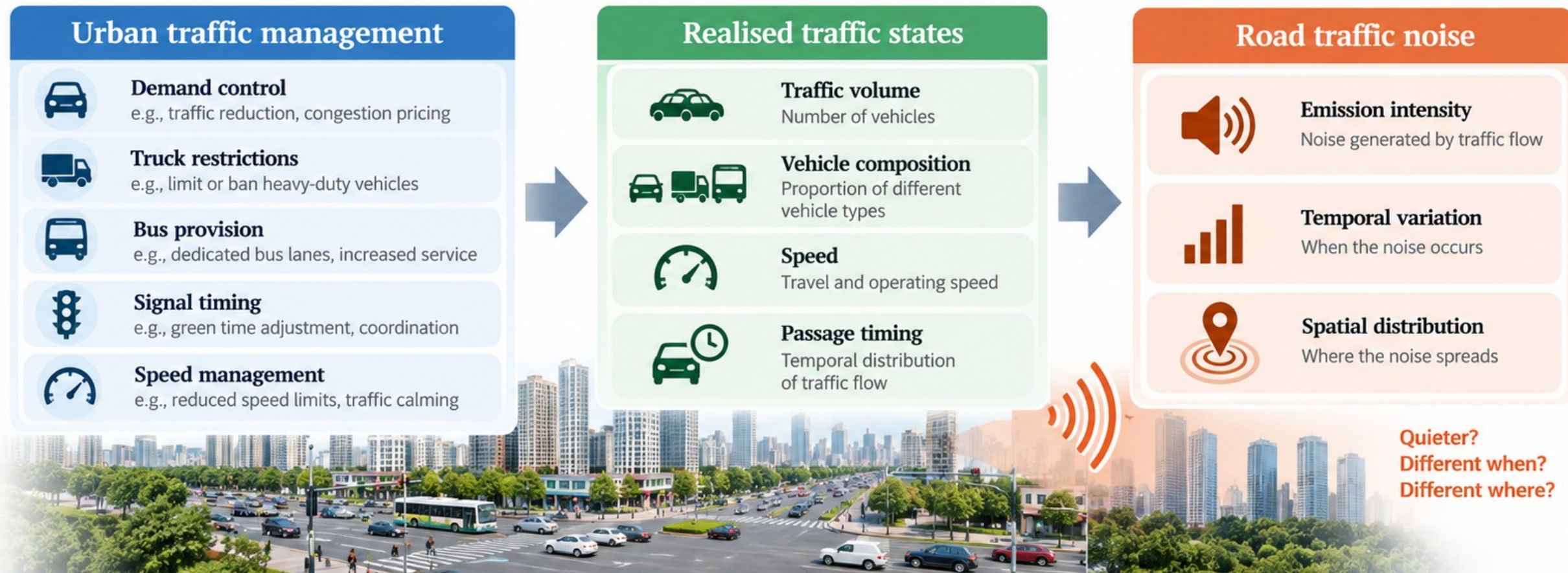
A coupled MATSim + NoiseModelling + Cesium Framework for Three Dimensional, MultiPeriod Simulation and Assessment of Traffic Control Noise Impacts

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South China University of Technology

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Traffic management reshapes urban traffic — and its noise emissions

Urban traffic management measures change not only the amount of traffic, but also its composition, speed and temporal patterns, which subsequently affect road traffic noise in magnitude, time and space.



Quieter?
Different when?
Different where?



Traffic management affects road traffic noise indirectly by changing the underlying traffic state.
From “How many dB?” to **why, when and where** noise changes occur.

What did we build?

1 Field observation & model validation

Collect real-world data and extract traffic and noise information to validate model parameters

Field Data Collection (Target Area)



Noise measurements

YOLO + DeepSORT (Computer vision)



-  Vehicle type classification
-  Traffic volume (flow)
-  Vehicle speed (per type)
-  Measured noise levels (LAeq, Lmax, etc.)

Traffic observations
(from video)

Noise observations
(from measurement)

MATSim Calibration / Validation

- Compare simulated and observed traffic volume
- Compare vehicle composition
- Compare vehicle speed and travel time
- Adjust MATSim parameters



NoiseModelling Calibration / Validation

- Compare modelled and measured noise levels (LAeq, spectral information, etc.)
- Adjust emission and propagation parameters



Validated model parameters

2 Scenario analysis and spatiotemporal noise assessment

From traffic-management scenarios to 3D visualization of noise impacts

Traffic-management scenarios

-  Demand adjustment
-  Truck restrictions
-  Bus provision
-  Signal control
-  Speed adjustment
- ... (87 scenarios in total)

MATSim Traffic simulation



Simulate traffic under different management scenarios

Realised traffic states (from MATSim)

-  Traffic volume
-  Vehicle composition
-  Vehicle speed
-  Passage timing (hourly)
-  Throughput / trip time
- ...

NoiseModelling Noise emission & propagation modelling



Calculate noise levels for each scenario and each hour

Noise results (from NoiseModelling)

-  Hourly sound levels (17 hours)
-  5,524 3D receivers
-  Scenario-based noise results (LAeq, Lden, etc.)

Cesium 3D visualization



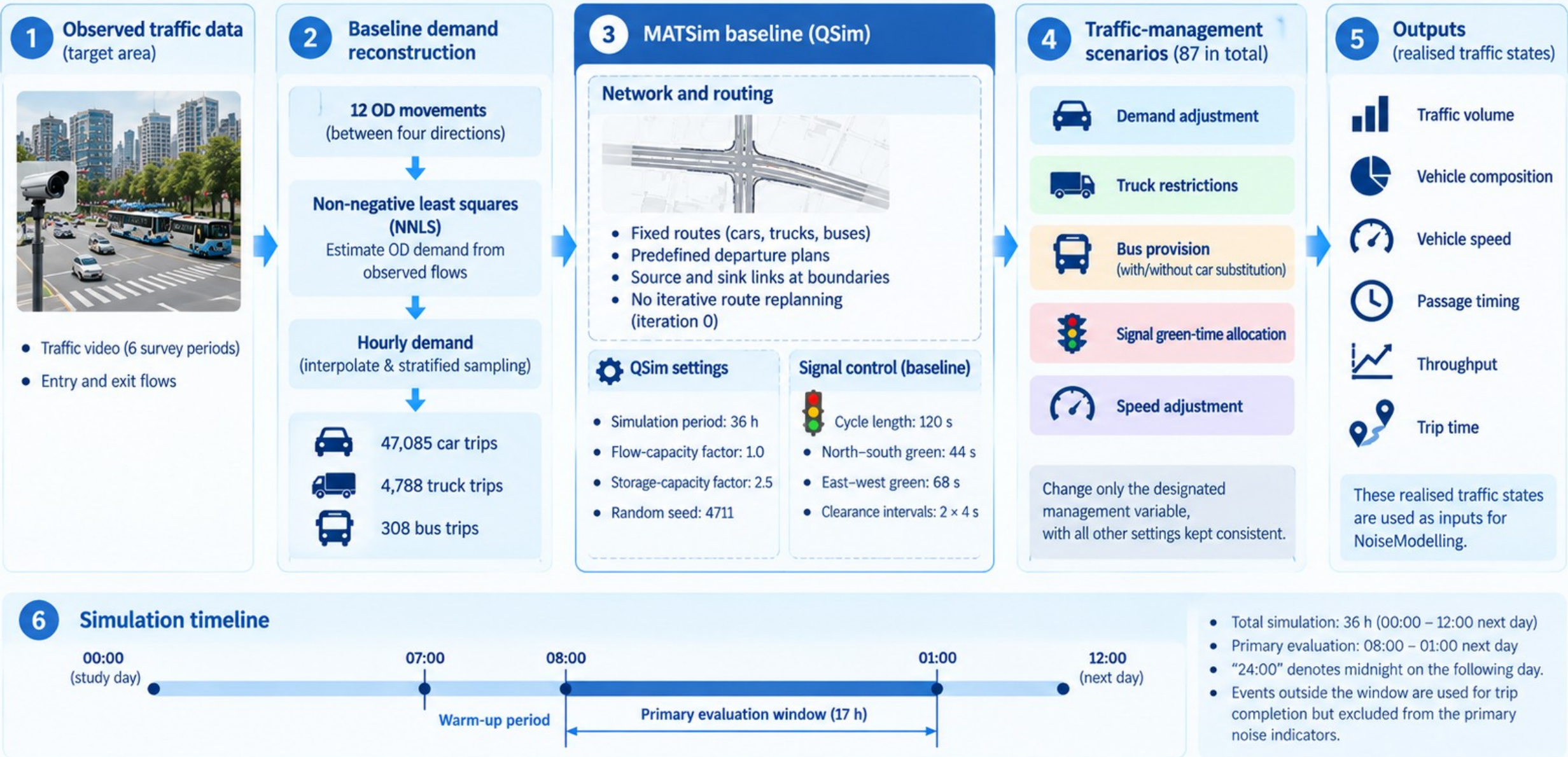
Visualize noise results in 3D urban environment

Spatiotemporal noise response



WHY?
WHEN?
WHERE?

MATSim — What happened to traffic?



NoiseModelling — What happened acoustically?

1 Traffic events from MATSim

Vehicle classes and link passage times



Vehicle ID	Type	Link ID	Entry time	Exit time
v1	car	L_101	08:12:03	08:12:15
v2	bus	L_102	08:15:20	08:15:47
v3	truck	L_101	08:16:10	08:16:28
...	...	...	...	...

3 Emission calculation

CNOSSOS-EU 2020

CNOSSOS-EU 2020 emission function

- Calculate sound power for each traversal (8 octave bands: 63 – 8,000 Hz)
- Assign each traversal's sound energy to the hour in which the vehicle left the link
- Pavement type: NL08
- Reference temperature: 20 °C
- Source height: 0.05 m
- Minimum speed: 20 km/h
- Statistical junction correction applied
- Idling and start-up acceleration not modelled explicitly

**Link-based, hour-dependent
sound power** (per vehicle class and octave band)

5 Propagation modelling

3D environment and receivers



- Road source (height: 0.05 m)
- Receivers (5,524)
- Buildings (3D)

Propagation settings

- Source search distance: 250 m
- Reflection search distance: 50 m
- Reflections: up to 3rd order
- Horizontal diffraction: enabled
- Vertical diffraction: disabled
- Common buildings, receivers and parameters across scenarios (reuse propagation matrix)

6 Acoustic outputs

Hourly, receiver-level and area-level results



Hourly sound levels

Hourly A-weighted equivalent levels at each receiver



Receiver-level results

5,524 valid receivers
(multi-height, 10 m horizontal spacing)



Area-level indicator

Energy-averaged across receivers and 17-hour window

What did we test?

1 Demand adjustment

Scale car and truck demand



- 10% – 200% of baseline (in 10% increments)
- Scale car and truck demand simultaneously
- Bus schedules unchanged

19 alternatives

(D010–D090, D110–D200)

2 Truck restrictions

Increase the restriction proportion



- 10% – 100% of trucks restricted (in 10% increments)
- Other traffic inputs unchanged

10 alternatives

(TR010–TR100)

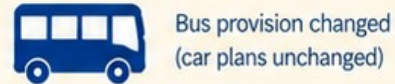
3 Bus provision

Adjust the number of scheduled bus trips

With car substitution



Without substitution

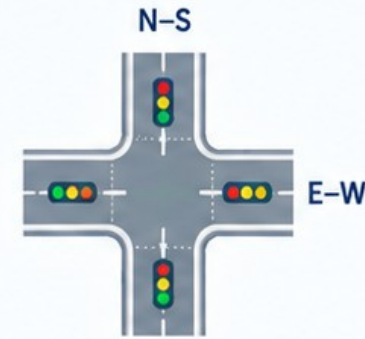


19 alternatives in each group

(B010–B090, B110–B200)

4 Directional capacity allocation

Reallocate signal green time



- 120-s cycle (two 4-s clearance intervals)
- N–S green time: 32–68 s (E–W adjusted accordingly)

6 alternatives

(G01–G06)

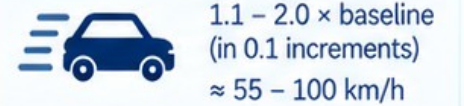
5 Speed adjustment

Change free-flow speed limits

Lower speed limits



Higher speeds (sensitivity)



- Use the smaller of original speed and the cap (V20–V40)
- Multiply baseline link speeds by 1.1–2.0 (S110–S200)

13 alternatives

(V20–V40, S110–S200)

Baseline scenario

(shared across all measures)

D100 = TR000 = B100 = G00 = V50



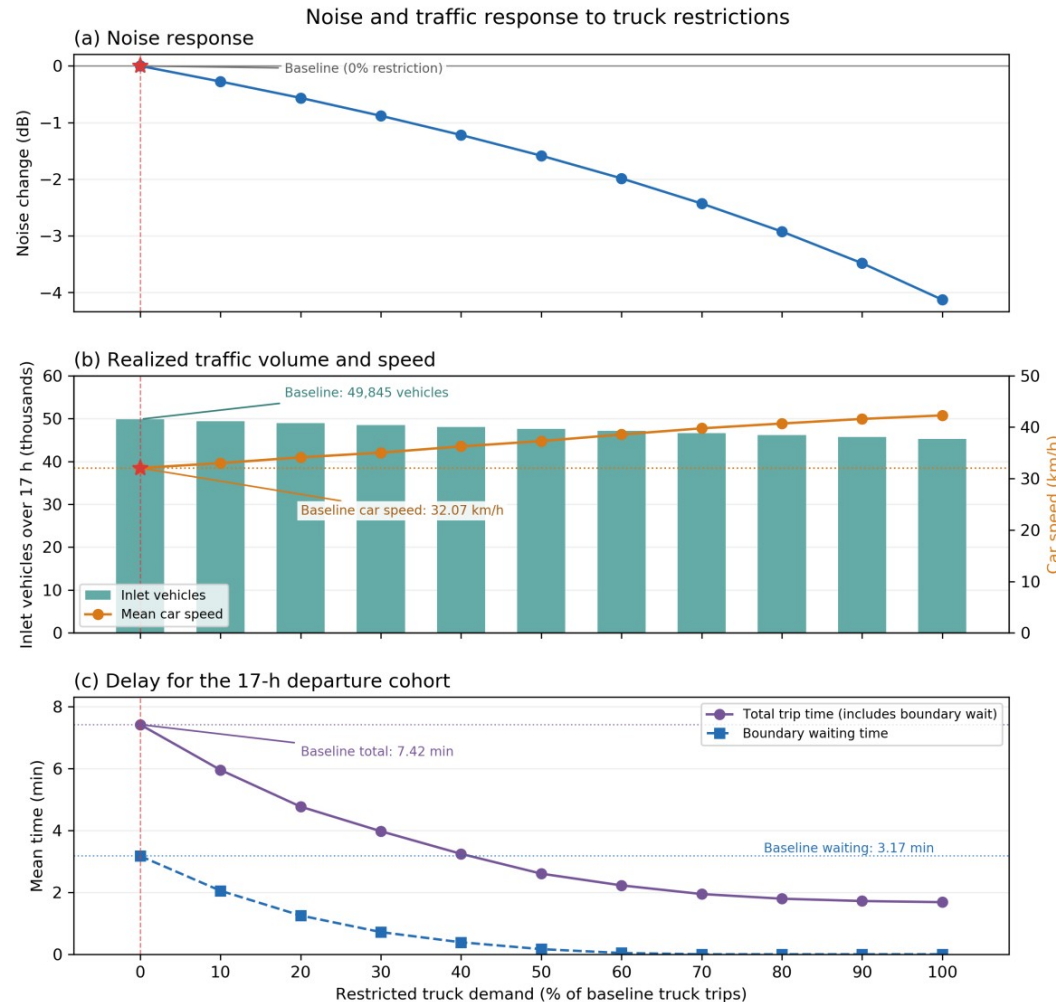
Total

87 unique scenarios

(1 baseline + 86 alternatives)

What did the traffic change and overall noise responses?

Hourly NoiseModelling outputs show patterns that a single aggregate indicator can't tell.



Demand / trucks

Persistent reductions are visible across much of the assessment period.

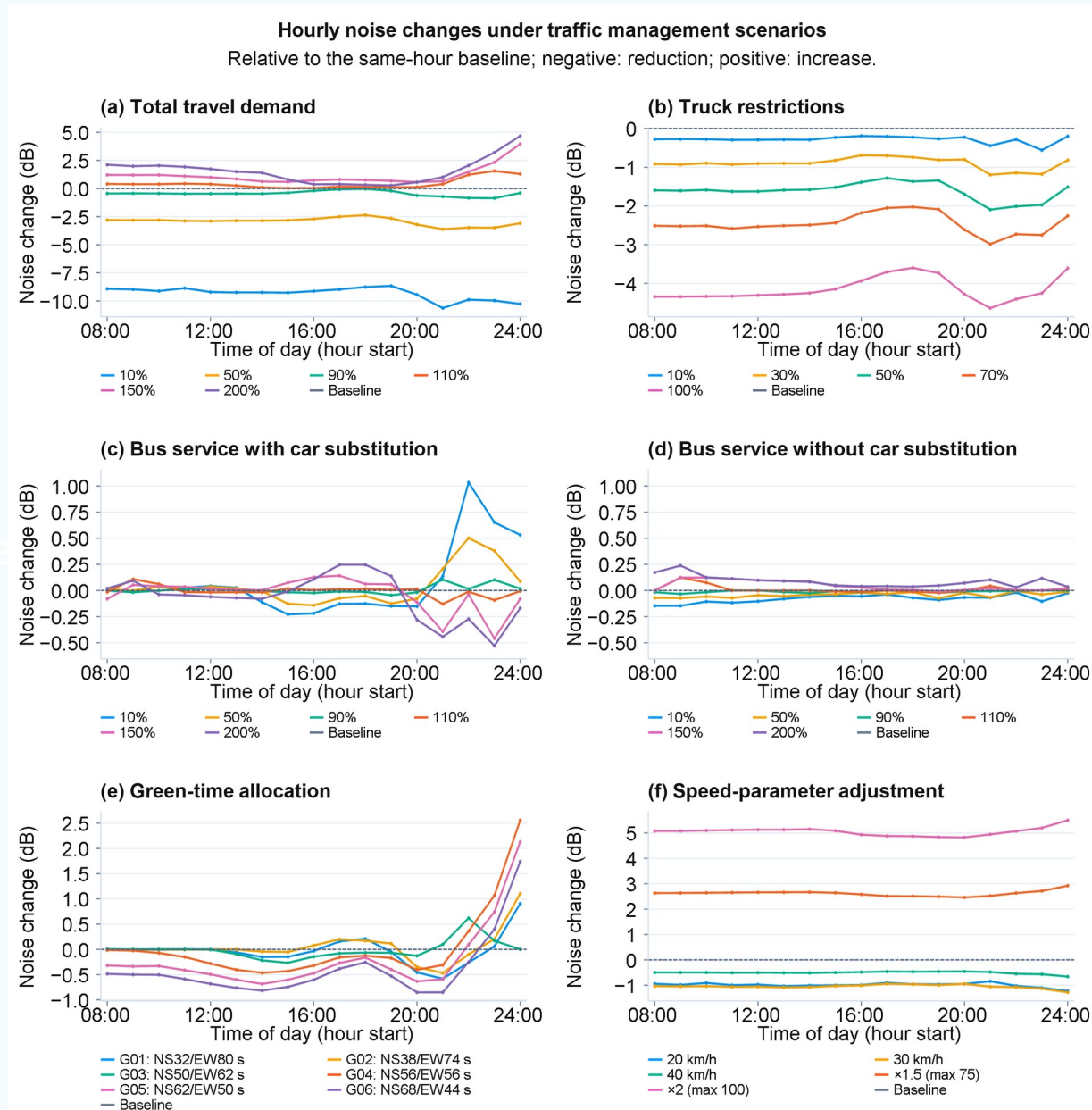
Signal timing

Some scenarios show later-hour increases despite lower average noise.

Interpretation

Average acoustic benefit can conceal temporal redistribution.

What did the workflow reveal over time?



Persistent reductions

Lower travel demand and greater truck removal consistently reduced noise across the modelled hours.

Time-dependent responses

Bus supply and green-time changes produced smaller, more variable effects, including increases during some late-evening hours.

Contrasting speed scenarios

Lower speed settings generally reduced noise, whereas higher free-speed settings increased it.

Key insight: Daily averages can mask hourly noise increases and differences in when benefits occur

What did the workflow reveal in 3D space?

Demand at 50%

Area: -2.882 dB | Median: -2.914 dB
Receivers with >1 dB reduction: 100%

Complete truck restriction

Area: -4.128 dB | Median: -3.986 dB
Receivers with >1 dB reduction: 100%

30 km/h speed limit

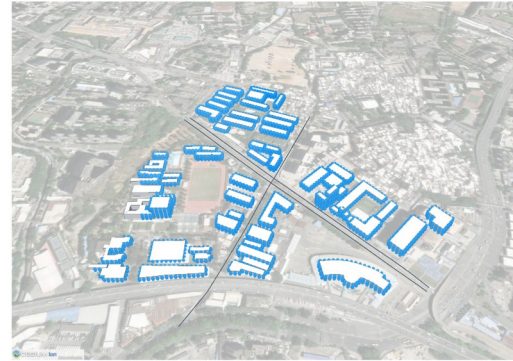
Area: -1.033 dB | Median: -1.045 dB
Receivers with >1 dB reduction: 51.09%



Building-facade noise differences with a shared colour scale

(a) Demand 50%

21:00–22:00 | Area change -3.615 dB



(b) Bus 50% with car substitution

22:00–23:00 | Area change 0.502 dB



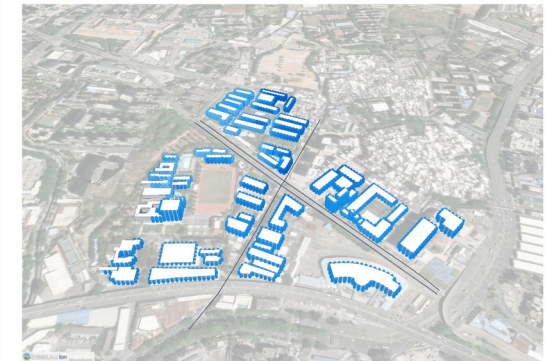
(c) Bus 50% without car substitution

19:00–20:00 | Area change -0.074 dB



(d) Truck removal 100%

21:00–22:00 | Area change -4.640 dB



(e) Green allocation G01

00:00–01:00 next day | Area change 0.907 dB



(f) Speed cap 30 km/h

00:00–01:00 next day | Area change -1.279 dB



What did we learn from the integrated workflow?

① Non-proportional response

Management intensity does not directly determine acoustic benefit.
→ Realised traffic states matters.

② Temporal redistribution

Average noise reduction can conceal delayed traffic activity.
→ Average benefit $\neq$ uniform temporal benefit

③ Spatial heterogeneity

Similar aggregate changes can produce different receiver-level patterns.
→ Δ dB alone does not describe spatial benefit

④ Traffic-acoustic mismatch

Noise reduction and traffic-performance improvement are not necessarily aligned.
→ Both outcomes should be evaluated together

Beyond “How many dB?” → Why did it change? When did it change? Where did it change?

Traffic management reshapes road traffic noise through changes in the underlying traffic state — not only in magnitude, but also in time and space.



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NoiseModelling Day

Thank You!

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